

Dedication: To all parents who have devoted themselves with careful attunement to raising securely attached children, which insures passage of secure attachment down through the generations; and with deep compassion, for all those parents who were not able to provide security of attachment, and for all the insecurely attached children of the world.

Chapter 1. Attachment Research: A History of Ideas

I. Maternal Deprivation

- In New York City 1931, the pediatric director at Bellvue Hospital became interested in the effects of deprivation of nurturing on infants in institutions. Babies had problems eating, sleeping and healing from infections ... developmental delays in speech and behavior ... some died despite medical care. "A pattern of physical, social, emotional, and cognitive impairment, and they became progressively more unresponsive."
- Disruptions in parental care in early childhood led to difficulty forming an attachment with foster or adoptive parents. Children became detached, indifferent towards caregivers and parents, only capable of superficial affection.
- In trying to understand these problems in institutionalized/foster care children, researchers began to focus not on separation from parents per se, but on the effects of maternal deprivation.
- (1951) John Bowlby consolidated research on maternal deprivation for the WHO and wrote "The quality of the parental care which the child receives in his earliest years is of vital importance for his future mental health."
- Bowlby on ideal care: "essential for mental health is that the infant and young child should experience a warm, intimate, and continuous relationship with his mother (or permanent mother-substitute)" -- too little of that experience constitutes maternal deprivation and adversely affects the child's developing mental health.
- In addition to deprivation due to separation, Bowlby highlighted a child living at home with a mother who is unable to provide sufficient care may also experience maternal deprivation.

II. Concept of Attachment Bonds

- Infants show instinctual behavioral patterns, for example reaching, following, crying, smiling, babbling and protesting when left alone. Bowlby coined the term "attachment behaviors" which are designed to attract the mother and keep the mother nearby.
- Attachment behaviour is any form of behaviour that results in a person attaining or maintaining proximity to some other clearly identified individual. It can be one or a few people, usually in order of preference.
- Attachment behavior is primarily associated with forming, maintaining, reacting to disruption of, and reestablishing attachment bonds.
- Bowlby suggested that the mother's availability and responsiveness are essential for the strengthening of the attachment bond. "If the infant seeks proximity and receives a welcoming or accepting response, then

attachment behavior is reinforced; in contrast, if the infant finds the mother to be repeatedly unresponsive, then the attachment behavior is deactivated."

Exploration

- The more an individual has a secure emotional bond with the caregiver, the more there is a natural tendency to explore progressively farther from the caregiver.
- Children with secure attachment bonds alternate between episodes of attachment behavior, such as proximity seeking, and exploratory behavior, progressing in the direction of increased exploration and autonomy.

III. Patterns of Attachment

- Influenced by personality theory, the idea that children derive a sense of emotional security from their parents, Mary Ainsworth developed the idea of the "secure base":
 - When a caregiver provides an infant with stable and consistent availability for protection, responsiveness, support, and comfort, the infant develops a sense of security in connection to that caregiver, and that security is the "base" from which the infant explores.
 - The greater the felt sense of security and confidence in the availability of the caregiver, the greater the infant's exploration.

Concept of Insecurely Attached Infants

- In Ainsworth's observations, some of the babies left unattended in their cribs for long periods of the day appeared non-attached. They failed to show crying and protest when the mother left or greeting behaviors when the mother returned. Ainsworth also noted that some of the babies cried excessively and were very clingy. She considered children who consistently showed these characteristics to be insecurely attached.

The Strange Situation

- Ainsworth hypothesized that differences in patterns of attachment would be amplified in an unfamiliar environment. When the mother is with the infant, attachment behavior would decrease and exploratory would increase. Conversely, when the mother leaves the room and returns, attachment behavior would increase and exploratory decrease.
- The Strange Situation (SS): One year-olds were observed with mother in an unfamiliar room with toys. Mother goes out of the room and then reunites.
- 3 groups were identified in the SS:
 - A "Avoidant" - characterized by avoidance of the mother, especially when the mother returned to the room. In contrast, when the mother returned to securely attached infants, they showed proximity and contact seeking behavior. Avoidant infants showed heightened exploratory and inhibited attachment behavior.

- B "Secure" - showed greater exploratory behavior when the mother was in the room, healthy protest when the mother left, greater proximity and contact seeking during reunion with the mother. Balanced interplay of exploratory and attachment behavior.
- C "Ambivalent" - less exploratory behavior. Cried significantly more when with mother, and during separation episodes. Extreme distress in the separation episodes. Moderate to strong seeking of proximity and to maintain contact once gained. Chronically anxious in relation to the mother, heightened attachment behavior and inhibited exploratory behavior.
- Ainsworth also examined whether the mother's behavior is associated with the 3 patterns of attachment:
 - Secure baby's mothers were more affectionate and tender when holding the babies, responded quicker to crying, and more sensitive, accepting, cooperative and accessible.
 - Avoidant baby's mothers were more rejecting, showed less positive feelings and more anger, less affectionate, greater aversion to physical contact.
 - Ambivalent baby's mothers showed unresponsiveness to crying, more involved in routine activities when holding the babies, insensitive to the infant's signals.
- Western studies show 65% - 75% secure in SS. Avoidant and ambivalent (13 - 18%). China and Japan distributions similar to Western. Varied percentages found in other cultures.

IV. Disorganized Attachment

- In early research with the SS, the babies that didn't fit the pattern of avoidant, ambivalent or secure were seen as unclassifiable and their data was ignored. Later, Mary Main established criteria for a fourth primary classification, which they called disorganized/disoriented (D).
- Such babies alternate between avoidant and ambivalent behaviors over time or show contradictory avoidant and ambivalent behaviors at the same time. In addition, during the SS such babies display misdirected and interrupted movements, mistimed and anomalous behaviors, freezing, pervasive indices of fear and apprehension, and outright signs of disorganization and disorientation.
- With respect to maternal behavior in the SS, mothers of disorganized babies were observed at times to be distinctly frightening to their children (e.g., looming in the child's face, talking too loudly, approaching the child too quickly or too suddenly, threatening the child, or being outright abusive). Additionally, these mothers at times appeared frightened of their own children (e.g., displaying deadpan expressionlessness, freezing behavior, and backing away from their own children in apparent fear).
- Main found that parents of disorganized children often have unresolved trauma or loss in their histories.

V. Stability of Attachment Patterns Over Time

- Patterns of attachment from 12 to 18 months are a property of the relationship with the caregiver. However, as the months pass, through the process of "internalization" and the linking of attachment behaviors with "internal working models" (discussed later), the inner organization of attachment becomes

ever more stable . . . over the course of childhood, adolescence and adulthood, and increasingly resistant to change over time.

- Alan Sroufe used the SS to study the stability of attachment patterns over time. He considered attachment behaviors observed in the SS as the manifestation of a “central organizing principle” in the development process of a child.

- The stability of attachment classification from 18 months to 20 years old: "as high as 72% to 77%".

VI. Attachment and the Organization of Development

- Minnesota Longitudinal Study (Sroufe): 179 young, expectant mothers, many came from homes of alcoholism, domestic violence, child abuse and maltreatment. Investigated the stability of attachment patterns over time, the effects of early attachment on personality development, and how parenting style affected attachment status.

- At 12 months 22% of the babies were resistant (Ainsworth's ambivalent), 20% avoidant.

- 18 months: secure children were less distressed by separation, 2 years: were more autonomous in exploratory behavior than 6 months earlier, 3 1/2 years; were more comfortable around peers than insecure children. A picture began to emerge how attachment patterns change form over time.

- Personality in preschool :

- Secure: greater self-esteem, more positive peer relationships than insecure children.

- Avoidant: bullies, shy loners or daydreamers.

- Ambivalent: impulsive, poor concentration / fearful, clingy.

- Securely attached children were likely to grow up into securely attached adults and insecure children were likely to retain their insecure status into adulthood unless they were exposed to healthy attachment figures.

- Insecure attachment correlated with later development of depression.

- Insecure children who were frequently distracted from exploratory play in the SS were more likely to show attention deficits.

- Ambivalent children were more likely to have an anxiety disorder as an adult.

- Disorganized attachment in infancy strongly correlated with psychopathology in early adulthood. Disorganized attachment in infancy aggravated by later sexual abuse correlated with self-injurious behavior.

VII. The Role of the Father

- Comparable rates of secure and insecure attachment to both mothers and fathers. But no correlation between the two was found (a toddler could show secure attachment towards one and not the other).
- Some research suggests different roles: mother's role more focused on proximity-seeking, father's role more focused on encouraging exploratory behavior.

VIII. Intergenerational Transmission

- Main found "Strong correlation between parent's pattern of attachment and the baby's."
- Bowlby concluded "across studies, the pattern of attachment of the parent on the AAI significantly predicted the pattern of attachment of his or her baby in the SS."
- Study of 3 generations: "Attachment status of the grandmother on the AAI significantly predicted the pattern of attachment of both the mother and also the mother's baby."

Reasons for Transmission

- May relate to metacognitive capacity (discussed later in the chapter)
- Unresolved, disoriented (related to group D):
 - In developing the Adult Attachment Interview (AAI), Main observed some adults showed disorganization around a specific trauma/loss and not others. They coined the new classification of "Unresolved, disoriented" (Ud) for discrete, temporary disorganization in response to specific AAI topics and inquiries. In contrast, adults with disorganized attachment (CC) show pervasive disorganization across the transcript.
 - Unresolved status for trauma and loss (Ud) in the parent was significantly associated with disorganized attachment in the child. Unresolved status effectively treated with psychotherapy correlated with secure children.
 - The cumulative effect of the parent's intense shifts in state and lapses toward disorganized states of mind is vulnerability of the child to developing multiple, contradictory internal working models (next section).

IX. The Organization of States of Mind: Internal Working Models (IWMs)

- Similar to how animal species develop a mental topographical map of their environment, Bowlby came to believe infants develop a map of their relational reality. Patterns of attachment can be explained by the operation of these internal maps of the quality of attachment with caregivers:
 - In the working model of the world that anyone builds, a key feature is the notion of who the attachment figures are, where they can be found, and how they're expected to respond. Similarly, in the working model of self, a key feature is the notion of how acceptable or unacceptable the person is him/herself in the eyes of his/her attachment figures.

- Based on these complementary models is the person's forecast of how accessible and responsive the attachment figures are likely to be should they be turned to for support.. confident they will be available, or afraid they will not be -- occasionally, frequently or most of the time.
- Patterns of attachment behavior can be explained by the operation of internal working models, which represent either secure or insecure experience with attachment figures.
- Through exposure to a healthy attachment figure in later childhood (after infancy), or an effective attachment-based therapy, an insecure IWM can be changed to a stable, secure one. Main refers to this as establishment of earned security.
- The IWM organizes and directs not only feelings and behavior, but also attention, memory, and cognition, to the extent that such mental functions are related to attachment.
- Main noticed IWMs are not only related to individual patterns of non-verbal behavior, but also to patterns of language and structure of mind.

X. Dynamic-Maturation Model of Attachment (DMM)

- Also interested in organization of mind in response to experience, Patricia Crittenden emphasized attachment-related patterns of information processing in response to experiences of danger or threat.
- When children experience danger or threat, strategies for coping develop over the course of childhood through an interplay of 1) maturation level at the time and 2) the degree to which attachment figures provide protection and comfort
- Distortions in the way information is processed to protect one's self from threat:
 - Cognitive strategy: Omitting affective information (group A avoidant) – omit feelings from processing and to act in accordance with expected consequences
 - Affective strategy: Omitting cognitive information (group C resistant) – Act in accordance with feelings with little attention to consequences. Process experience according to its intensity and the resulting autonomic nervous system responses.
 - Disorganized: Uses both cognitive and affective strategies, but in alternating, less integrated ways (A/C) or in a blended way (AC) that integrates, to varying degrees, “false, denied, and delusional affect and cognition.”
 - Secure: balanced integration of cognitive and affective information, in an integrated way that does not distort or omit any information.
- Every strategy is considered to “reflect learned patterns of mentally managing cognitive and affective information so as to protect and adapt to dangerous circumstances”
- The particular ways that experience is processed lead to specific strategies for coping, for “eliciting needed caregiving from the parent[,] and for reducing possible rejection or harm.”

- Though adaptive from this perspective, any tendency to privilege or exclude available information will distort information in ways that often lead to heightened expectation of danger and, thus, to the use of “self-protected behavior under safe circumstances”, which is characteristic of insecure attachment.

XI. Identification of States of Mind

- Main used information processing concepts to differentiate attachment types in the Adult Attachment Interview (AAI). Emphasized coherence of mind, for example secure adults describe early attachment relationships in an organized, coherent manner.
- Overall, dismissing, preoccupied, and disorganized adults show low coherence of discourse in describing their early attachment experiences on the standardized Adult Attachment Interview.
- Discovery that attachment patterns are represented by “states of mind” inferred through discourse allows researchers to explore older children’s and adults’ representations of attachment.

XII. Three Generations of Metacognition

- Defined as direct perception of one’s own state of mind (any cognitive or affective experience).
- Main discovered using the AAI that secure adults tended to have a greater capacity to reflect on their attachment experiences.
- AAI transcripts characterized by frequent examples of metacognitive statements are likely to be secure transcripts. For example “It *seemed* as if my mother was angry when I was growing up,”
- The London School (2nd Gen) advanced metacognition and attachment through work on *mentalization* – the knowledge of one’s own and other’s mental states as mental states. Reflective function (RF) is the capacity to mentalize.
 - The RF scale ranges from -1 to 9. Patients with a personality disorder or dissociative disorder scored less than 3. Mentalization based treatment (MBT) causes many patients to no longer meet diagnostic criteria for a personality disorder.
- The Rome School (3rd Gen) took a modular view of metacognition, identified 4 distinct metacognitive abilities. Discovered that low scores in specific categories coincided with specific mental disorders.
- Treatment can focus on specific metacognitive deficits in a patient with a specific mental disorder.

XIII. Intersubjectivity

- Discovery that maternal responsiveness predicts secure attachment in the SS laid the foundation of research on parental attunement. Researchers began to understand that the *interactions* between mother and child as a secure or insecure dyad were of equal significance to maternal responsiveness.

- Donald Winnicott presented the idea that holding, physically and emotionally, conveys a careful attunement by a mother toward her infant, and allows the child to develop a physical representation of bodily experience and also a representation of mind. He also suggested play is a central ingredient to development of a sense of *self*. Encouraging and reassuring of spontaneous expressions likely leads to a healthy and stable sense of self.
- The “still face” experiment strongly illustrated the importance of the reciprocal nature of infant–caregiver attunement and the profound negative effects of even a brief misattunement in an otherwise secure infant–mother dyad.
- Mutually regulated communication and cooperation is characterized by mutual imitative behavior, emotional empathy, and reciprocal nonverbal and verbal communication. By 12 months, the baby discovers his or her own mind and the mind of the other, and also that subjective experiences can be communicated and shared.
- Attachment and intersubjectivity are separate and complementary motivational systems. The attachment system balances our related needs for the security of physical proximity and the learning that exploration makes possible. The intersubjective system is driven by our need to know and be known by others. If attachment exists to foster felt security, intersubjectivity exists to promote the experience of psychic intimacy and belonging.
- Well-synchronized mother–infant interactions predict secure attachment.

XIV. Attunement, Misattunement, and Synchronization of Expression

- Beatrice Beebe’s research focused on forms of intersubjectivity -- synchronized, coordinated timing of interactional patterns and rhythms between infant and mother . . . exchanges such as gaze, gesture, touch, and vocal pitch.
- The opposite of these synchronized interactions is misattunement. Circumstances of misattunement in insecure infants:
 - Mothers of ambivalent babies were misattuned to a spectrum of the baby’s expressions, but were “extremely attentive and well attuned when the baby was afraid.”
 - Mothers of avoidant babies were misattuned to expressions of negative feelings, especially toward them, or when seeking comfort or reassurance. However they excelled in attuning to exuberance of babies when mastering a new toy or game.
- Beebe researched intersubjectivity in treating adult patients. She made use of nonverbal communication to mirror the patient’s verbal rhythms, facial expressions, and body language.
- Dolores, a patient with intense fear, withdrawal and dissociation initially could not look at Beebe. Dolores gradually came to sense a “comforted” inner state as she became more aware of how Beebe matched her and coordinated with her. Coordination simultaneously gave Beebe a greater feeling of “being with” Dolores.